RANDOM FOREST MACHINE LEARNING PROJECT

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Project Title: Customer Churn Prediction Using Random Forest Classifier

Dataset Name: Telco Customer Churn (Kaggle – WA_Fn-UseC_-Telco-Customer-Churn.csv). **Dataset size:** 7,043 rows and 21 columns. **Source:** https://www.kaggle.com/datasets/blastchar/telco-customer-churn

Problem Statement

Telecommunication companies may lose revenue when customers discontinue their services. The objective of this project is to build a classification model that predicts whether a customer will leave the company (Churn = Yes) or stay (Churn = No), using customer, service, contract and billing information.

Data Preprocessing

The dataset was explored using head(), info(), describe() and value_counts(). TotalCharges was converted from text to numeric; its 11 missing values were filled using the median. The customerID column was removed because it is an identifier. The target Churn was encoded as No = 0 and Yes = 1. Categorical input features were converted using one-hot encoding (pd.get_dummies with drop_first=True). The resulting feature matrix contained 30 columns. The data was split into 5,634 training records (80%) and 1,409 testing records (20%).

Model Used

A RandomForestClassifier with 100 decision trees (n_estimators=100) and random_state=42 was trained on the training data. The model was then used to predict churn for the test data.

Evaluation Metrics – Original Random Forest Model

Metric	Result
Accuracy	78.92%
Precision (Churn = Yes)	0.643
Recall (Churn = Yes)	0.458
F1-score (Churn = Yes)	0.535
Confusion Matrix	TN = 941, FP = 95, FN = 202, TP = 171

Bonus Work: GridSearchCV tested n_estimators = [100, 200] and max_depth = [10, 20] with 3-fold cross-validation and F1 scoring. The best parameters were max_depth = 10 and n_estimators = 100; best cross-validation F1-score = 0.567. The tuned model achieved accuracy = 81.41%, precision = 0.698, recall = 0.525 and F1-score = 0.599 on the same test set. Five-fold cross-validation of the original model gave an average accuracy of 78.93%.

Key Findings

The three most important features in the original model were TotalCharges (0.190), tenure (0.176) and MonthlyCharges (0.172). The tuned model performed better than the original model on all four reported test metrics. However, the original model's churn recall of 0.458 shows that it missed a number of customers who actually churned.

Conclusion

The Random Forest Classifier was used to predict customer churn from the Telco Customer Churn dataset. Data cleaning, median imputation, categorical encoding and an 80:20 train-test split were completed. The tuned model improved test accuracy to 81.41% and F1-score to 0.599. The results suggest that tenure and billing-related features are useful for churn prediction. The model can support customer-retention planning, although further improvement in identifying customers who churn is still possible.